# Alphabets, strings, and languages

## Computation and Discrete Structures III

**Juan Marcos Caicedo Mejía – jmcaicedo@icesi.edu.co**

Department of Computing and Intelligent Systems
Barberi Faculty of Engineering, Design and Applied Sciences
ICESI University

*Material adapted from Professor Oscar Bedoya*

2026

# Contents


## 1. Alphabets, words and languages

1.1 Overview
1.2 Alphabet
1.3 Strings (Words)
1.4 Languages

## 2. Operators on words and languages

2.1 Operations in Strings
2.2 Operations in Languages

# Contenido


## 1. Alphabets, words and languages

1.1 Overview
1.2 Alphabet
1.3 Strings (Words)
1.4 Languages

## 2. Operators on words and languages

# Contenido


## 1. Alphabets, words and languages

1.1 Overview
1.2 Alphabet
1.3 Strings (Words)
1.4 Languages

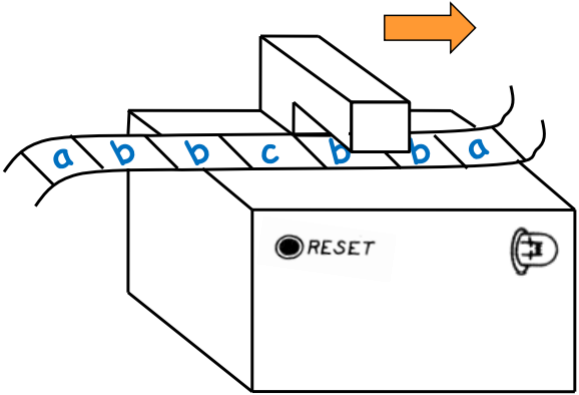
## 2. Operators on words and languages

2.1 Operations in Strings
2.2 Operations in Languages

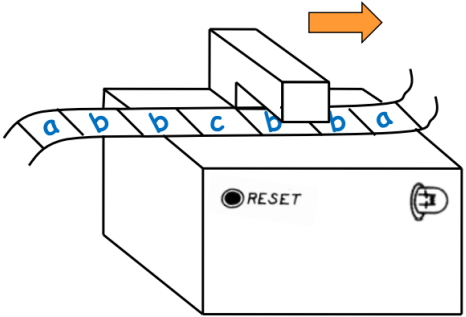
# Languages, Machine Types and Grammars

| Type | Languages | Machine Type | Grammar Rules |
|---|---|---|---|
| 0 | Recursively enumerable | Turing machine | Unrestricted |
| 1 | Context-sensitive | Linear bounded automaton | $\alpha \to \beta, \quad \lvert\alpha\rvert \leq \lvert\beta\rvert$ |
| 2 | Context-free | Pushdown automaton | $A \to \gamma$ |
| 3 | Regular | Finite automaton | $A \to aB$<br>$A \to a$ |

# Theoretical machine (tape)



# Theoretical machine (tape)




The **alphabet** is the **set of symbols** that may appear in the input of the machine

# Contenido


## 1. Alphabets, words and languages

1.1 Overview
1.2 Alphabet
1.3 Strings (Words)
1.4 Languages

## 2. Operators on words and languages

2.1 Operations in Strings
2.2 Operations in Languages

# Alphabets

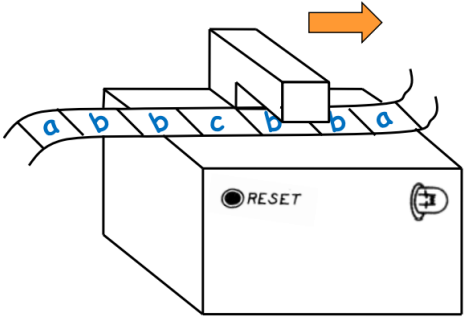
## Alphabet

An alphabet is any non-empty set of symbols

- $\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- $\Sigma = \{a,b\}$

- Latin alphabet:
  $\Sigma = \{a,b,c,d,e,f,g,h,i,j,k,l,m,n,ñ,o,p,q,r,s,t,u,v,w,x,y,z\}$
- Greek alphabet:
  $\Sigma = \{\alpha, \beta, \gamma, \delta, \epsilon, ..., \Psi, \Omega\}$

# Words or strings




**Words** or **strings** are finite sequences of symbols

# Contenido


## 1. Alphabets, words and languages

1.1 Overview
1.2 Alphabet
1.3 Strings (Words)
1.4 Languages

## 2. Operators on words and languages

2.1 Operations in Strings
2.2 Operations in Languages

# Notion of word

Given the alphabet used in Spanish:
$\Sigma = \{a,b,c,d,e,f,g,h,i,j,k,l,m,n,ñ,o,p,q,r,s,t,u,v,w,x,y,z\}$
Words can be created:

- Colina
- Puente
- Dardo
- Fdkfjk

The notion of word has no associated semantics

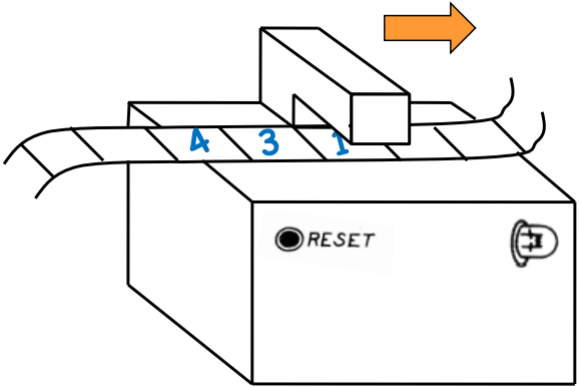
# Strings or words

## String or word

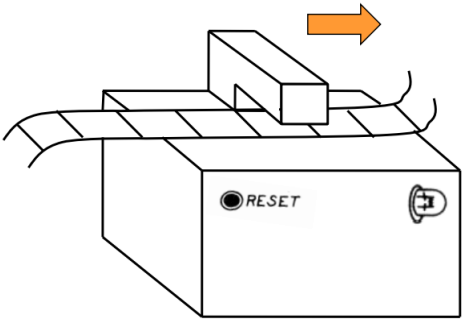
A word is a finite sequence of symbols from a given alphabet

- If $\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, then 431, 021, $\epsilon$, are words over $\Sigma$
- If $\Sigma = \{a,b\}$ then ab, ba, aaab, $\epsilon$, are words over $\Sigma$

# Theoretical machine (tape) with word

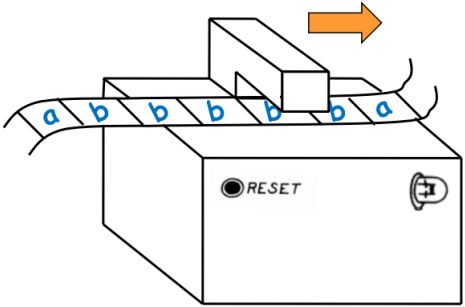


# Machine (tape) with empty string




The **empty string** $\epsilon$ represents a word that has 0 symbols, that is, an empty tape

# Machine (tape) that accepts strings




A machine **accepts** a specific set of words that can be generated from an alphabet

# Contenido


## 1. Alphabets, words and languages

1.1 Overview
1.2 Alphabet
1.3 Strings (Words)
1.4 Languages

## 2. Operators on words and languages

2.1 Operations in Strings
2.2 Operations in Languages

# Language

## Language

A language is a particular set of words

# Language

## Language

A language is a particular set of words

- Show the following languages defined over $\Sigma = \{a,b\}$

# Language

## Language

A language is a particular set of words

- Show the following languages defined over $\Sigma = \{a,b\}$
  - L1: set of words that have exactly 3 symbols

# Language

## Language

A language is a particular set of words

- Show the following languages defined over $\Sigma = \{a,b\}$
  - L1: set of words that have exactly 3 symbols
  - L2: set of words that have at least one a

# Language

## Language

A language is a particular set of words

- Show the following languages defined over $\Sigma = \{a,b\}$
  - L1: set of words that have exactly 3 symbols
  - L2: set of words that have at least one a
  - L3: set of words that have an even number of symbols

# Language

## Language

A language is a particular set of words

- Show the following languages defined over $\Sigma = \{a,b\}$
  - L1: set of words that have exactly 3 symbols
  - L2: set of words that have at least one a
  - L3: set of words that have an even number of symbols
  - L4: set of all possible words

# Universal language

## Universal language over $\Sigma$

- It is denoted as $\Sigma^*$ and is also known as the **closure**
- $\Sigma^*$ is the language formed by all strings over the alphabet $\Sigma$

# Universal language

## Universal language over $\Sigma$

- It is denoted as $\Sigma^*$ and is also known as the **closure**
- $\Sigma^*$ is the language formed by all strings over the alphabet $\Sigma$

- Show the universal language $\Sigma^*$ for the following alphabets:
  - $\Sigma = \{a,b,c\}$
  - $\Sigma = \{1\}$

# Universal language

## Universal language over $\Sigma$

- $\Sigma^*$ is the language formed by all strings over the alphabet $\Sigma$

- For $\Sigma = \{a,b,c\}$, $\Sigma^* = \{\epsilon, a, b, c, aa, ab, ac, ba, bb, bc, \ldots\}$
- For $\Sigma = \{1\}$, $\Sigma^* = \{\epsilon, 1, 11, 111, 1111, \ldots\}$

# Universal language

## Universal language over $\Sigma$

- $\Sigma^*$ is the language formed by all strings over the alphabet $\Sigma$

- For $\Sigma = \{a,b,c\}$, $\Sigma^* = \{\epsilon, a, b, c, aa, ab, ac, ba, bb, bc, \ldots\}$
- For $\Sigma = \{1\}$, $\Sigma^* = \{\epsilon, 1, 11, 111, 1111, \ldots\}$

## Membership of the empty string and infiniteness of the universal language

- $\epsilon$ is always in $\Sigma^*$ because the empty string can be obtained from any alphabet
- For any alphabet $\Sigma$, $\Sigma^*$ is infinite because $\Sigma$ cannot be empty

# Language

## Language

- A language $L$ over an alphabet $\Sigma$ is a subset of $\Sigma^*$, that is, $L \subseteq \Sigma^*$

# Contenido

1. Alphabets, words and languages

2. Operators on words and languages
   2.1 Operations in Strings
   2.2 Operations in Languages

# Contenido


1. Alphabets, words and languages
   1.1 Overview
   1.2 Alphabet
   1.3 Strings (Words)
   1.4 Languages

2. Operators on words and languages
   2.1 Operations in Strings
   2.2 Operations in Languages

# Concatenation of strings

## Concatenation $\cdot$

Let $x$ and $y$ be two strings, their concatenation is the string $x \cdot y$

- $a \cdot aba = aaba$
- $ab \cdot ba = abba$

# Power of a string

## Power of a string

Let $x$ be a string, the power of $x$ is defined as:

$$x^n = \begin{cases} \epsilon & n = 0 \\ x \cdot x^{n-1} & n \geq 1 \end{cases}$$

$\cdot$ is the concatenation operator

# Power of a string

## Power of a string

Let $x$ be a string, the power of $x$ is defined as:

$$x^n = \begin{cases} \epsilon & n = 0 \\ x \cdot x^{n-1} & n \geq 1 \end{cases}$$

$\cdot$ is the concatenation operator

- $(aab)^3 = aab \cdot (aab)^2$
  $= aab \cdot aab \cdot aab^1$
  $= aab \cdot aab \cdot aab \cdot aab^0$
  $= aab \cdot aab \cdot aab \cdot \epsilon = aabaabaab$

# Power of a string

## Power of a string

Let $x$ be a string, the power of $x$ is defined as:

$$x^n = \begin{cases} \epsilon & n = 0 \\ x \cdot x^{n-1} & n \geq 1 \end{cases}$$

Show:

- $a^3 \cdot (aba)^2$
- $(ab)^2 \cdot (ba)^3$

# Examples of languages

- $\Sigma = \{a\}$,
  $L = \{a, aa, aaa, aaaa, ...\}$

# Examples of languages

- $\Sigma = \{a\}$,
  $L = \{a, aa, aaa, aaaa, ...\} = \{a^n \mid n \geq 1\}$

# Examples of languages

- $\Sigma = \{a\}$,
  $L = \{a, aa, aaa, aaaa, ...\} = \{a^n \mid n \geq 1\}$, strings of one or more a's

# Examples of languages

- $\Sigma = \{a\}$,
  $L = \{a, aa, aaa, aaaa, ...\} = \{a^n \mid n \geq 1\}$, strings of one or more a's
- $\Sigma = \{a, b\}$,
  $L = \{ab, aabb, aaabbb, ...\}$

# Examples of languages

- $\Sigma = \{a\}$,
  $L = \{a, aa, aaa, aaaa, ...\} = \{a^n \mid n \geq 1\}$, strings of one or more a's
- $\Sigma = \{a, b\}$,
  $L = \{ab, aabb, aaabbb, ...\} = \{a^n b^n \mid n \geq 1\}$

# Examples of languages

- $\Sigma = \{a\}$,
  $L = \{a, aa, aaa, aaaa, ...\} = \{a^n \mid n \geq 1\}$, strings of one or more a's
- $\Sigma = \{a, b\}$,
  $L = \{ab, aabb, aaabbb, ...\} = \{a^n b^n \mid n \geq 1\}$, strings with the same number of a's as b's, where the a's are to the left of the b's

# Examples of languages

- $\Sigma = \{a\}$,
  $L = \{a, aa, aaa, aaaa, ...\} = \{a^n \mid n \geq 1\}$, strings of one or more a's
- $\Sigma = \{a, b\}$,
  $L = \{ab, aabb, aaabbb, ...\} = \{a^n b^n \mid n \geq 1\}$, strings with the same number of a's as b's, where the a's are to the left of the b's
- $\Sigma = \{0, 1\}$,
  $L = \{\epsilon, 01, 10, 0011, 0101, 1100, 1001, ...\}$

# Examples of languages

- $\Sigma = \{a\}$,
  $L = \{a, aa, aaa, aaaa, ...\} = \{a^n \mid n \geq 1\}$, strings of one or more a's
- $\Sigma = \{a, b\}$,
  $L = \{ab, aabb, aaabbb, ...\} = \{a^n b^n \mid n \geq 1\}$, strings with the same number of a's as b's, where the a's are to the left of the b's
- $\Sigma = \{0, 1\}$,
  $L = \{\epsilon, 01, 10, 0011, 0101, 1100, 1001, ...\} = \{w \in \{0, 1\}^* \mid$
  have the same number of 0's as 1's$\}$

# Examples of languages

- $\Sigma = \{a\}$,
  $L = \{a, aa, aaa, aaaa, ...\} = \{a^n \mid n \geq 1\}$, strings of one or more a's
- $\Sigma = \{a, b\}$,
  $L = \{ab, aabb, aaabbb, ...\} = \{a^n b^n \mid n \geq 1\}$, strings with the same number of a's as b's, where the a's are to the left of the b's
- $\Sigma = \{0, 1\}$,
  $L = \{\epsilon, 01, 10, 0011, 0101, 1100, 1001, ...\} = \{w \in \{0, 1\}^* \mid$
  have the same number of 0's as 1's$\}$, strings with the same number of 0's as 1's

# Length of a string

## Length of a string

Let $x$ be a string that belongs to a language $L$, its length is denoted by $|x|$ and defined as:

$$|x| = \begin{cases} 0 & x = \epsilon \\ n & x = a_1 a_2 \ldots a_n \end{cases}$$

# Length of a string

## Length of a string

Let $x$ be a string that belongs to a language $L$, its length is denoted by $|x|$ and defined as:

$$|x| = \begin{cases} 0 & x = \epsilon \\ n & x = a_1 a_2 \ldots a_n \end{cases}$$

- $|\epsilon| = 0$
- $|ababaa| = 6$

# Contenido


1. Alphabets, words and languages
   1.1 Overview
   1.2 Alphabet
   1.3 Strings (Words)
   1.4 Languages

2. Operators on words and languages
   2.1 Operations in Strings
   2.2 Operations in Languages

# Concatenation of languages

## Concatenation of languages

Let $A$ and $B$ be two languages defined over $\Sigma$, the concatenation $A \cdot B$ is defined as:

$$A \cdot B = \{u \cdot v \mid u \in A \wedge v \in B\}$$

- $A = \{a, ab, ac\}, B = \{b, b^2\}$
- $A \cdot B =$

# Concatenation of languages

## Concatenation of languages

Let $A$ and $B$ be two languages defined over $\Sigma$, the concatenation $A \cdot B$ is defined as:

$$A \cdot B = \{u \cdot v \mid u \in A \wedge v \in B\}$$

- $A = \{a, ab, ac\}, B = \{b, b^2\}$
- $A \cdot B = \{ab, abb, acb, ab^2, abb^2, acb^2\} = \{ab, ab^2, acb, ab^2, ab^3, acb^2\} = \{ab, acb, ab^2, ab^3, acb^2\}$

# Power of a language

## Power of a language

Given a language $A$ over $\Sigma$ the power is defined as:

$$A^n = \begin{cases} \{\epsilon\} & n = 0 \\ A \cdot A^{n-1} & n \geq 1 \end{cases}$$

Compute $A^3$ for $A = \{ab, b\}$

# Power of a language

## Power of a language

Given a language $A$ over $\Sigma$ the power is defined as:

$$A^n = \begin{cases} \{\epsilon\} & n = 0 \\ A \cdot A^{n-1} & n \geq 1 \end{cases}$$

Compute $A^3$ for $A = \{ab, b\}$,
$A^3 = A \cdot A \cdot A \cdot A^0 = \{ab, b\}\{ab, b\}\{ab, b\}\{\epsilon\} = \{ab, b\}\{abab, bab, abb, bb\} =$
$\{ababab, abbab, ababb, abbb, babab, bbab, babb, bbb\} \cdot \{\epsilon\}$

# Exercises: power of languages

Given $A = \{ab, ca, ad\}$,

- Is $abcaab \in A^3$?
- Is $adca \in A^2$?
- Is $caba \in A^2$?
- Is $abcaaa \in A^3$?
- Is $adcaab \in A^3$?

# Exercises: power of languages

Given $A = \{ab, ca, ad\}$,

- Is $abcaab \in A^3$? Yes
- Is $adca \in A^2$? Yes
- Is $caba \in A^2$? No
- Is $abcaaa \in A^3$? No
- Is $adcaab \in A^3$? Yes

# Exercises: power of languages

Given $A = \{ab, c, ac\}$,

- Is $accab \in A^3$?
- Is $abacca \in A^3$?
- Is $abcc \in A^3$?
- Is $abcba \in A^3$?

# Exercises: power of languages

Given $A = \{ab, c, ac\}$,

- Is $accab \in A^3$? Yes
- Is $abacca \in A^3$? No
- Is $abcc \in A^3$? Yes
- Is $abcba \in A^3$? No

# Kleene closure

## Kleene closure

The Kleene closure of a language $A$ is the union of its powers, denoted by $A^*$

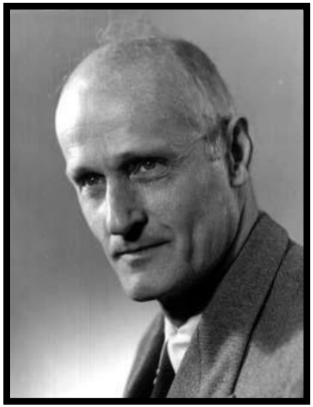
$$A^* = A^0 \cup A^1 \cup A^2 \cup \ldots$$

- It is also known as **Kleene star / Star closure**
- $A^*$ is the set of possible concatenations over $A$

# Stephen Kleene

**Stephen Kleene**

- Creator of regular expressions
- Introduced Kleene closure, $A^*$



(1909 - 1994)

# Kleene closure

Compute $A^*$ for $A = \{a, ab\}$

# Kleene closure

$A = \{a, ab\}$

- $A^0 = \{\epsilon\}$
- $A^1 = \{a, ab\}$
- $A^2 = \{aa, aab, aba, abab\}$
- ...

$A^* = \{\epsilon, a, ab, aa, aab, aba, abab, ...\}$

# Kleene closure

$A = \{a, ab\}$

- $A^0 = \{\epsilon\}$
- $A^1 = \{a, ab\}$
- $A^2 = \{aa, aab, aba, abab\}$
- ...

$A^* = \{\epsilon, a, ab, aa, aab, aba, abab, ...\}$

- Is $ababab \in A^*$?
- Is $abbbbb \in A^*$?
- Is $abaaaaaaa \in A^*$?

# Kleene closure

$A = \{a, ab\}$

- $A^0 = \{\epsilon\}$
- $A^1 = \{a, ab\}$
- $A^2 = \{aa, aab, aba, abab\}$
- ...

$A^* = \{\epsilon, a, ab, aa, aab, aba, abab, ...\}$

- Is $ababab \in A^*$? Yes
- Is $abbbbb \in A^*$? No
- Is $abaaaaaaa \in A^*$? Yes

# Kleene closure

$A = \{a, ab\}$

- $A^0 = \{\epsilon\}$
- $A^1 = \{a, ab\}$
- $A^2 = \{aa, aab, aba, abab\}$
- ...

$A^* = \{\epsilon, a, ab, aa, aab, aba, abab, ...\}$

## Relationship between $A^*$ and $\Sigma^*$

What relationship does $A^*$ have with $\Sigma^*$?
Compute $\Sigma^*$ over $\Sigma = \{a, b\}$

# Kleene closure

$A = \{a, ab\}$

- $A^0 = \{\epsilon\}$
- $A^1 = \{a, ab\}$
- $A^2 = \{aa, aab, aba, abab\}$
- ...

$A^* = \{\epsilon, a, ab, aa, aab, aba, abab, ...\}$
$\Sigma^* = \{\epsilon, a, b, aa, ab, ba, bb, aaa, aab, ...\}$

# Kleene closure $A^*$ and $\Sigma^*$ closure

## Closures

- $\Sigma^*$ is defined over the alphabet and corresponds to all strings that can be created over an alphabet $\Sigma$
- $A^*$ is defined over a language $A$ and consists of all possible concatenations

# Kleene closure $A^*$ and $\Sigma^*$ closure

## Closures

- $\Sigma^*$ is defined over the alphabet and corresponds to all strings that can be created over an alphabet $\Sigma$
- $A^*$ is defined over a language $A$ and consists of all possible concatenations

$A = \{a, ab\}$ is defined over $\Sigma = \{a, b\}$

- $A^* = \{\epsilon, a, ab, aa, aab, aba, abab, ...\}$
- $\Sigma^* = \{\epsilon, a, b, aa, bb, ab, ba, aaa, aab, aba, ...\}$
- **In general,** $A^* \subseteq \Sigma^*$ **holds**

# Kleene positive closure

## Kleene positive closure $A^+$

The Kleene positive closure of a language $A$ is the union of the powers excluding $A^0 = \{\epsilon\}$,

$$A^+ = A^1 \cup A^2 \cup A^3 \cup \ldots$$

# Positive closure $\Sigma^+$

## Positive closure $\Sigma^+$

It is the set of words that can be formed over $\Sigma$ excluding the empty string $\epsilon$

# Exercises – Closures

- Let $A = \{a, b, ab\}$, show $A^*$ and $A^+$. Indicate if $abba \in A^*$, $bbaa \in A^*$
- Let $A = \{a, aa, ac\}$ and $B = \{b, ba\}$, show $A \cdot B$, $B \cdot A$ and $B^*$
- Let $A = \{a\}$, show $A^*$

# Exercises – Closures

- **Let** $A = \{a, b, ab\}$**, show** $A^*$ **and** $A^+$**. Indicate if** $abba \in A^*$**,** $bbaa \in A^*$
  - $A^* = A^0 \cup A^1 \cup A^2 \cup \ldots$
    $= \{\epsilon\} \cup \{a, b, ab\} \cup \{aa, ab, aab, ba, bb, bab, aba, abb, abab\} \cup \ldots$
    $= \{\epsilon, a, b, ab, aa, ab, aab, ba, bb, bab, aba, abb, abab, \ldots\}$
  - $A^+ = A^1 \cup A^2 \cup \ldots$
    $= \{a, b, ab\} \cup \{aa, ab, aab, ba, bb, bab, aba, abb, abab\} \cup \ldots$
    $= \{a, b, ab, aa, ab, aab, ba, bb, bab, aba, abb, abab, \ldots\}$
- **Let** $A = \{a, aa, ac\}$ **and** $B = \{b, ba\}$**, show** $A \cdot B$**,** $B \cdot A$ **and** $B^*$
  - $A \cdot B = \{ab, aba, aab, aaba, acb, acba\}$
  - $B \cdot A = \{ba, baa, bac, baa, baaa, baac\}$
  - $B^* = \{\epsilon, b, ba, bba, bab, bbba, babb, \ldots\}$
- **Let** $A = \{a\}$**, show** $A^*$
  - $A^* = \{\epsilon, a, aa, aaa, aaaa, aaaaa, \ldots\}$

# Exercises – Languages

Show strings that belong to the following languages. Indicate if the empty string $\epsilon$ belongs to the languages and express in general terms (in words) the type of strings that belong to each one.

- $L_1 = \{w_1 c w_2 \mid |w_1| = |w_2| \text{ where } w_1, w_2 \in \Sigma^* \text{ with } \Sigma = \{a, b\}\}$
- $L_2 = \{a^n b^m \mid n \neq m, n, m \geq 0\}$
- $L_3 = \{a^n b^{2n} c^n \mid n \geq 0\}$

# Exercises – Languages

$L_1 = \{w_1 c w_2 \mid |w_1| = |w_2| \textbf{ where } w_1, w_2 \in \Sigma^* \textbf{ with } \Sigma = \{a, b\}\}$
$aca, acb, bca, abbbabcaaaaaa$
In general, strings that have a c in the middle, such that the substrings on either side have the same length. $\epsilon \notin L_1$

# Exercises – Languages

$L_2 = \{a^n b^m \mid n \neq m, n, m \geq 0\}$
$abb, aab, aabbb, aaabb$
In general, strings that have a different number of a's than b's where the a's are to the left of the b's. $\epsilon \notin L_2$

# Exercises – Languages

$L_3 = \{a^n b^{2n} c^n \mid n \geq 0\}$
$abbc, aabbbbcc, aaabbbbbbccc$
In general, strings that have twice as many b's as a's and c's where a's, b's, and then c's appear from left to right. $\epsilon \in L_3$

# Exercises – Languages

Express formally the following languages:

- $L_1$ is the set of strings from the universal language of $\Sigma = \{a, b, c\}$ that start with a and end with a
- $L_2$ is the set of strings with even length defined over the universal language of $\Sigma = \{a, b\}$

# Exercises – Languages

- $L_1$ **is the set of strings from the universal language of** $\Sigma = \{a, b, c\}$ **that start with a and end with a**
  - $L_1 = \{aw_1a \mid w_1 \in \Sigma^* \text{ with } \Sigma = \{a, b, c\}\}$
- $L_2$ **is the set of strings with even length defined over the universal language of** $\Sigma = \{a, b\}$
  - $L_2 = \{w_i \mid |w_i| = 2k, \text{where there exists } k \geq 1, w_i \in \Sigma^* \text{ with } \Sigma = \{a, b\}\}$

# References

Kozen, D. C. (2007)
Automata and computability
*Springer Science & Business Media*. Lectures 1–2.